

인공지능수학기초 (01)

Mid-term Exam

Apr. 22, 2025

1. (4 pts) A, B , 그리고 X 가 모두 다 $n \times n$ 행렬일 때, 다음 방정식에서 X 를 구하시오.

$$A(XB)^{-1} = (B^{-1}A^2)^2$$

2. (6+4 pts) 아래 연립 선형 방정식을 지정하는 방법으로 푸시오.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} \& \underline{\mathbf{b}} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}$$

- (a) LU factorization을 이용한 방법
(b) Cramer's rule.

3. (3+3 pts) 다음 행렬의 column space와 null space의 basis를 각각 구하시오.

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 2 \\ 0 & 1 & 1 & -1 & 3 \end{pmatrix}.$$

4. (4+4+4 pts) 다음 행렬 A 를 보자.

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}.$$

- (a) A 의 eigenvalue와 eigenvector를 구하시오.
(b) A 를 대각화 (diagonalization) 하시오.
(c) A^{2025} 을 구하시오.

5. (4+4 pts) 다음 문장이 맞으면 증명하고, 틀리면 반례를 드시오.

- (a) 행렬 A 가 대각화 가능하면, $\det(A)$ 는 모든 eigenvalue들의 곱과 같다.
(b) $n \times n$ 행렬 A 와 B 가 둘 다 rank가 n 이면, AB 의 rank도 n 이다.

$$1. \quad A(XB)^{-1} = (B^{-1}A^2)^{-2}$$

$$AB^{-1}X^{-1} = B^{-1}A^2B^{-1}A^2$$

$$AB^{-1} = B^{-1}A^2B^{-1}A^2X$$

$$\therefore X = (B^{-1}A^2B^{-1}A^2)^{-1}AB^{-1}$$

$$= A^{-2}BA^{-2}BAB^{-1}$$

$$2. a) U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow 2 \text{ րդ}$$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow 2 \text{ րդ}$$

$$I U \underline{x} = \underline{b}, \quad U \underline{x} = \underline{y} \quad \& \quad I \underline{y} = \underline{b}$$

$$\underline{y} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} \rightarrow 1 \text{ րդ} \quad \& \quad \underline{x} = \begin{pmatrix} -1 \\ -2 \\ 3 \end{pmatrix} \rightarrow 1 \text{ րդ}$$

$$b) \det(A) = -2$$

$$\det(A_1) = 2$$

$$\det(A_2) = 4$$

$$\det(A_3) = -6$$

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$$3. A \rightarrow \begin{pmatrix} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a) basis of column space

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

b) basis of null space

$$x_3 = s, x_4 = t.$$

$$x_5 = 0, x_2 + x_3 - x_4 = 0$$

$$x_2 = -s + t$$

$$x_1 - x_3 + x_4 = 0$$

$$x_1 = x_3 - x_4 = s - t$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 5-t \\ -5+t \\ 5 \\ t \\ 0 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$\therefore$ basis of null space

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$4. (A - \lambda I) = \begin{pmatrix} -\lambda & 1 \\ 2 & 1-\lambda \end{pmatrix}$$

$$|A - \lambda I| = \lambda(\lambda-1) - 2 = \lambda^2 - \lambda - 2 = (\lambda-2)(\lambda+1)$$

$$a) \quad i) \lambda = -1, \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$ii) \lambda = 2, \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\therefore -1, \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad 2, \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

1 step

$$b) \quad S = \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix} \quad \& \quad \Lambda = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

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$$S = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} \quad \& \quad \Lambda = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$

$$c) A = S \Lambda S^{-1} \quad \therefore A^{2025} = S \Lambda^{2025} S^{-1}$$

$$\begin{aligned} \therefore A^{2025} &= \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} (-1)^{2025} & 0 \\ 0 & 2^{2025} \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 2^{2025} \end{pmatrix} \frac{1}{-3} \begin{pmatrix} 2 & -1 \\ -1 & -1 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2^{2025} - 2 & 2^{2025} + 1 \\ 2^{2026} + 2 & 2^{2026} - 1 \end{pmatrix} \end{aligned}$$

5. (a) 대각화 가능 $A \sim \Lambda$,

$$\text{i.e. } \Lambda = S^{-1} A S.$$

$$\begin{aligned} \therefore \det(\Lambda) &= \det(S^{-1} A S) \\ &= \det(S^{-1}) \det(A) \det(S) \\ &= \det(A) \end{aligned}$$

$$\& \det(\Lambda) = \prod_{i=1}^n \lambda_{ii} \quad (\lambda_{ii} \text{은 eigenvalue } \frac{\mathbb{Z}}{2})$$

$$\therefore \det(A) = \text{모든 eigenvalue } \frac{\mathbb{Z}}{2} \text{의 곱과 같다.}$$

$$b) \text{rank}(A) = \text{rank}(B) = n.$$

$$\Leftrightarrow \det(A) \neq 0 \& \det(B) \neq 0.$$

$$\therefore \det(AB) = \underbrace{\det(A)}_{\neq 0} \underbrace{\det(B)}_{\neq 0} \neq 0$$

$$\therefore \text{rank}(AB) = n.$$

